Pimpri Chinchawad
COLLEGE OF ENGINEERING
Akurdi, Pune

Presentation
On
Subject: UI and UX Design
by

Ms. Rohini S. Tambe
Ms. Deepali Jawale
Dr. Anagha Chaudhari
Department of Computer Engineering

What is Prototyping in UI/UX?

Prototyping is the process of creating a **visual and interactive representation** of a digital product (website/app). It simulates user interactions like clicking, scrolling, navigation, etc.

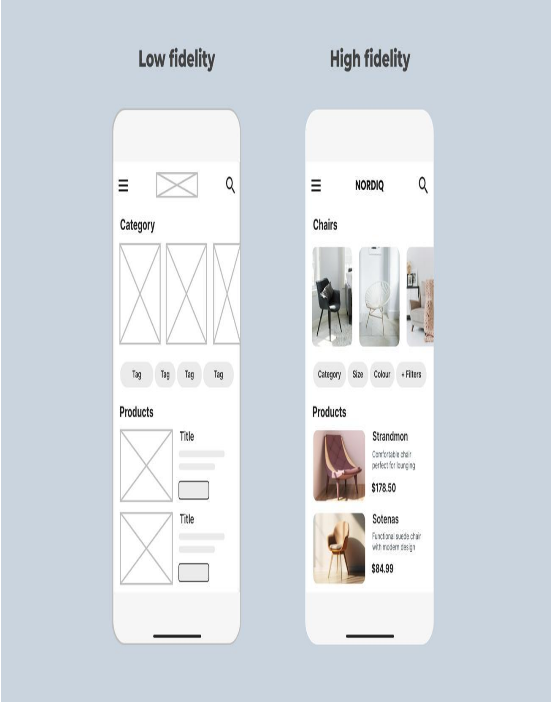
👉 It ranges from **simple sketches (low-fidelity)** to **fully interactive designs (high-fidelity)**.

Types :

1. Low fidelity
2. Mid Fedility
3. High Fedility

LOW_MID_HIGH


Profile
ratul sarkar
24
A Subtitle
project name
Profile
Ratul Sarkar
UX designer
Daily progress
Current Projects
Ecommerce app
Profile
Ratul Sarkar
UX designer
Edit profile
24
Ongoing
14
Inprogress
08
Completed
02
Rejected
Daily progress
Current Projects
Ecommerce app

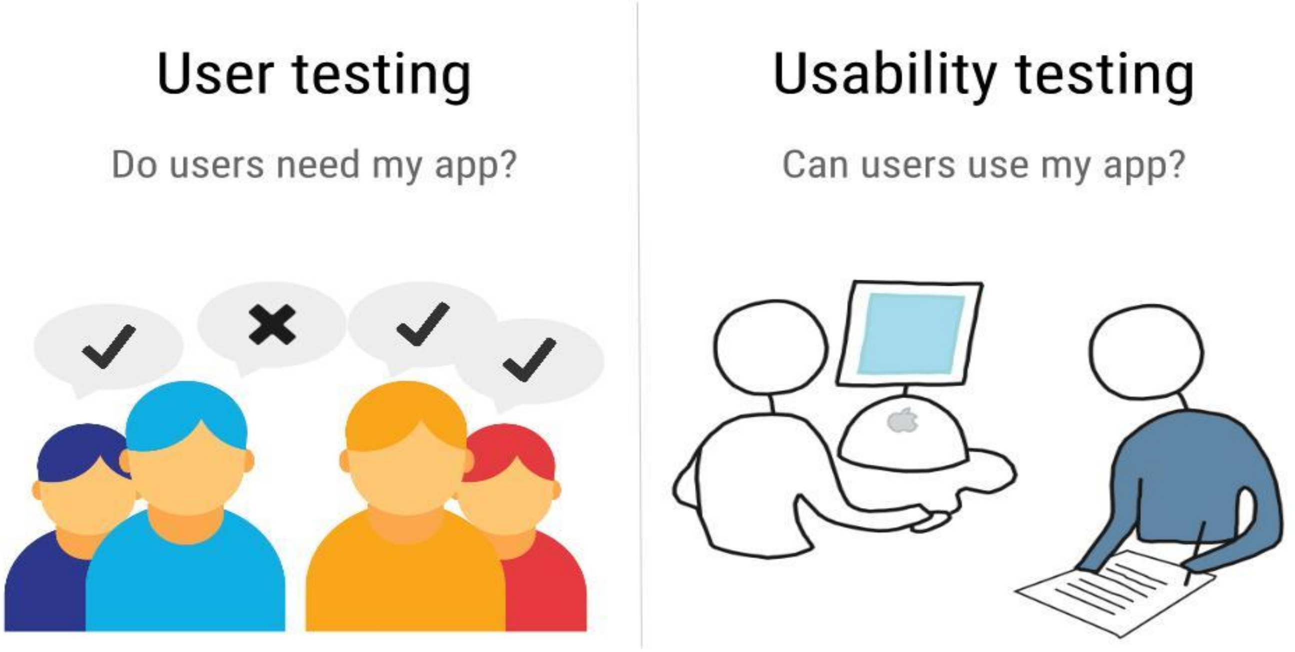

Low fidelity
High fidelity
Category
Tag
Tag
Tag
Tag
Products
Title
Title
NORDIQ
Chairs
Category
Size
Colour
+ Filters
Products
Strandmon
Comfortable chair perfect for lounging
$178.50
Sotenas
Functional suede chair with modern design
$84.99

What is Usability Testing?

Usability Testing is a method used in UI/UX design to evaluate how easily real users can use a product (website/app).
It involves observing users as they perform tasks to identify problems, confusion, or inefficiencies.

👉 Goal: Ensure the product is **easy, efficient, and satisfying to use**
Example: Food Delivery App

User is given a task:
👉 “Order a pizza”



What is Usability Testing

Usability testing is a technique used in UI/UX design to evaluate how easy and user-friendly a product (website, app, or system) is by testing it with real users.

It involves observing users as they try to complete tasks on your product to identify:

- Confusion points
- Errors
- Navigation issues
- User satisfaction

What is Usability Testing

Objectives of Usability Testing

- Improve **ease of use**
- Identify **design flaws**
- Enhance **user satisfaction**
- Validate **design decisions**
- Reduce development costs (by fixing issues early)

Types of Usability Testing

1. Moderated Testing
2. Unmoderated Testing
3. Remote Testing
4. In-Person Testing

Types of Usability Testing

1. Moderated Testing

- Conducted with a **facilitator (UX researcher)**
- Researcher asks questions during testing

👉 Example:
Researcher asks: *“What are you thinking while selecting this option?”*

2. Unmoderated Testing

- No facilitator
- User completes tasks independently

👉 Example:
User records screen while using an app at home

Types of Usability Testing

3. Remote Testing

- Conducted online (Zoom, tools)
- Users from different locations

👉 Example:
Testing an app with users from different cities

4. In-Person Testing

- Conducted in lab or office
- Direct observation

👉 Example:
User comes to lab and performs tasks under observation

Types of Usability Testing

5. Explorative Testing

- Done in early stage
- Open-ended feedback

👉 Example:
"Explore this app and tell what you feel"

6. Comparative Testing

- Compare two designs

👉 Example:
Version A vs Version B of homepage

Methods of Usability Testing

1. Think-Aloud Method

- User speaks thoughts while using system

👉 Example:
"I don't understand this button…"

✔ Helps understand user confusion

2. Task-Based Testing

- Users perform specific tasks

👉 Example:

- Login
- Add item to cart
- Checkout

✔ Measures success rate

Methods of Usability Testing

3. A/B Testing

- Two versions shown to different users

👉 Example:

- Red button vs Blue button

✔ Finds better design

4. Eye Tracking

- Tracks where user looks

👉 Example:

- User ignores important button

✔ Improves layout

Heuristic Evaluation (Heuristic Testing)

Introduction

- Heuristic Evaluation is a **usability inspection method**
- UI is evaluated using **predefined usability principles**
- Conducted by **UX experts (not real users)**
- Introduced by **Jakob Nielsen & Rolf Molich (1990)**

Key Idea

- Experts check interface against **Nielsen's 10 Heuristics**
- Helps identify **usability issues early**
- **Low cost & fast method**
- No need for user testing

Why Heuristic Evaluation?

- Detect usability problems early
- Improve user experience
- Reduce development cost
- Quick evaluation process
- Works well with prototypes

Heuristic Evaluation

Definition:
Heuristic Evaluation is a **usability inspection method** where experts evaluate a user interface using **Nielsen's 10 usability principles (heuristics)**.

Key Points:

- Conducted by **3–5 UX experts**
- No real users involved
- Focuses on identifying **usability problems**
- Each issue is mapped to a **specific heuristic**
- Issues are given a **severity rating (0–4)**

Simple Real-Life Example

Case: Login Page of a Mobile App

Scenario

A user tries to log in but:

- Enters wrong password
- Clicks login
- Nothing happens (no message shown)

Step-by-Step Heuristic Testing

1. Identify the Problem

- No feedback after clicking login

2. Map to Heuristic

- Violates: **Visibility of System Status**
  👉 System should inform user what is happening

3. Assign Severity

- Severity: **3 (Major Problem)**
  👉 User is confused and cannot proceed

4. Suggest Improvement

- Show message:
  “Incorrect password. Try again.”
- Add loading indicator

Visibility of System Status

- System should give **feedback to users**
- Example: Loading spinner, progress bar
- ❌ No response after clicking button

Match with Real World

- Use **simple, user-friendly language**
- Example: Shopping cart icon
- ❌ Technical terms like “ERR_404”

User Control & Freedom

- Provide **undo, redo, cancel options**
- Users should easily exit mistakes
- ❌ No back or cancel option
-

Consistency & Standards

- Maintain **uniform design & terminology**
- Example: Same icon for same action
- ❌ Different labels for same function

Error Prevention

- Prevent errors before they happen
- Example: Disable invalid inputs
- ❌ Allow wrong data submissio

Recognition vs Recall

- Reduce memory load
- Show options instead of remembering
- Example: Menus, suggestions
- ❌ Command-based systems

Flexibility & Efficiency

- Support both **beginners & experts**
- Example: Keyboard shortcuts
- ❌ No shortcuts or customizatio

Minimalist Design

- Keep interface **clean & simple**
- Show only relevant info
- ❌ Cluttered dashboards

H9 – Error Handling

- Show clear error messages
- Suggest solutions
- Example: “Password must be 8 characters”
- ❌ “Error occurred”

H10 – Help & Documentation

- Provide help when needed
- Example: Tooltips, FAQs
- ❌ No guidance for users

Severity Rating Scale

- 0 – No problem
- 1 – Cosmetic
- 2 – Minor issue
- 3 – Major issue
- 4 – Critical problem

Slide 17: Evaluation Process

1. Briefing
2. Independent evaluation
3. Record issues
4. Discussion (debrief)
5. Final report

Slide 18: Team Size

- Recommended: **3–5 evaluators**
- 5 evaluators find **~75% issues**
- 1 evaluator finds only **~35% issues**

Slide 19: Example Issue

- Issue: No feedback on login button
- Heuristic: Visibility of system status
- Severity: 3 (Major)
- Fix: Add loading indicator

Slide 20: Advantages

- Fast and inexpensive
- No users required
- Works early in design
- Easy to perform

Slide 21: Limitations

- Depends on expert knowledge
- May miss real user problems
- Not a substitute for usability testing

Slide 22: Heuristic vs Usability Testing

Heuristic Evaluation	Usability Testing
Experts	Real users
Fast	Time-consuming
Low cost	Higher cost
Finds guideline issues	Finds real behavior issues

Nielsen’s 10 Heuristics (Overview)

1. Visibility of system status
2. Match with real world
3. User control & freedom
4. Consistency & standards
5. Error prevention
6. Recognition over recall
7. Flexibility & efficiency
8. Aesthetic & minimalist design
9. Error recovery
10. Help & documentation